Explaining Human-perceived Difficulty of Chess Positions: Survey and Preliminary Experiments

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ABSTRACT

In this paper we survey existing approaches for estimating difficulty of chess positions as perceived by human players. Additionally, we offer our own implementation of rating prediction on chess puzzles using hand-crafted features, primarily based on the properties of the search tree and on performance of weak engines. We explain both classifiers and regressors using SHAP and LIME.

KEYWORDS

chess, explainability, classification, regression, feature engineering

1 INTRODUCTION

Explaining why a given chess position is perceived as easy or hard has been a long-standing goal both for human players and artificial intelligence researchers. This can be seen as a subset of the problem of explaining game states in general zero-sum two-player games. Possible applications of a solution to this problem range from tutoring software to automatic skill estimation in said games.

In this paper we overview existing approaches in this domain, as well as offer our own implementation of puzzle difficulty prediction. This problem could be formulated in many different ways, and could easily be seen as an interdisciplinary endeavor, however in this work we take a primarily machine learning view. In particular, we think of difficulty estimation as either a regression problem, where given an arbitrary chess position, we simply wish to predict its rating, or a a classification problem, where we should first bin all available positions into classes, say, *easy*, *medium*, and *hard*.

A possible issue with such approaches is the apparent subjectivity of rating assignment, however it will become clear that the process is not unlike assigning ratings to human players and adjusting them based on tournament, or individual game performances, which seems to be a generally accepted method of estimating players' skill level.

Another possible approach would be to avoid numerical ratings altogether, and instead treat difficulty estimation as purely a ranking problem. In this paradigm, the model should simply state which of the given two (or more) positions is perceived as more difficult. However, it is clear that this does not eliminate subjectivity whatsoever.

In this paper we implement new features and combine them with existing ones, with the goal of explaining difficulty of chess puzzles using LIME and SHAP. Furthermore we demonstrate the effectiveness of state-of-the-art tabular data models on a challenging task of difficulty estimation.

2 RELATED WORK

In recent years we have observed an increased interest in the problem of predicting puzzle difficulty, largely due to the IEEE BigData 2024 Cup [42] and the FedCSIS 2025 knowledgepit.ai Competition [41], both of which focused precisely on this topic.

The best performing models in both cases belonged either to the deep learning paradigm, e.g. [22, 28], or to the gradient boosting family, e.g. [3]. Interestingly, even the former approaches often utilized hand-crafted features, which is seldom seen in other fields, however it is a good indication of the difficulty of this problem.

A number of hand-crafted features are also described in [6, 7, 35]. Here the idea is to use the breadth and depth of the search tree explored by the engine as a proxy for the difficulty of the given position. For this approach to work, the pruning should coincide with the human thinking process, e.g. checks, and captures should be considered no matter what, while certain nonsensical moves can be rejected instantly without further search.

2.1 Human-play Modelling

Since chess engines started surpassing the skill of human players by a large margin, attention has somewhat shifted to developing engines that exhibit a more human playing style with the purpose of making the moves more intuitively understantable. This is by no means a solved problem, as it is difficult to even quantify what constitutes a human-like playing style.

Arguably the first big breakthrough in this regard was the AlphaZero engine [34], which was trained purely via self-play, and was claimed to produce more human-like moves compared to other state-of-the-art engines at the time. This claim, as well as the possible drawbacks, were discussed, for instance, in [5].

Other noteworthy human-modelling engines come from the Maia family, namely Maia-1 [21], Maia-2 [37], and Maia-3 [23]. The latter, in particular, simultaneously models human play, maximizes playing strength and offers interpretability.

ALLIE [40] is yet another recent attempt at modelling human play, with particular emphasis on predicting the next move based on the history of preceding moves. Similar approach can be seen in Otter [31], but with additional capability of modelling clock management.

2.2 Machine Learning for Tabular Data

Recently, substantial progress has been made in the field of machine learning for tabular data, which can be seen for instance on the TabArena benchmark [12]. Recent foundational models such as Kumo-Tabular [24] and TabFM [19] have surpassed previous state-of-the-art models such as XGBoost [8], CatBoost [10], LightGBM [18], and their variants, all based on the gradient boosting paradigm. Importantly, these models work out-of-the-box, i.e. require no additional domain-specific fine-tuning.

2.3 Engine Strength Estimation

The problem of assigning a rating to chess engines has been of interest even before their level of play was comparable to that of a human [2, 39]. The initial belief was that an estimated Elo rating grows linearly with the depth of engine's search capabilities, i. e. that adding a ply of depth increases the engine's Elo by a constant. Due to hardware limitations, diminishing returns of increasing search depth were only observed later, e.g. in [17].

Currently, engines are evaluated according to the Elo rating system, with rating changing based on tournament play, e.g. [9, 16, 32]. However, engine tournaments are not universally standardized, and can use different time controls, hardware, etc. which means that ratings from one engine to another are not necessarily comparable. They are certainly not directly comparable with human ratings either, despite using the same rating system.

2.4 Explainability

Chess has proven to be a relevant and relatively challenging domain for Explainable AI (XAI). A lengthy survey of recent approaches can be found in [4], for a more general XAI survey, see [33].

State-of-the-art machine learning models, including those discussed in Section 2.2, typically do not offer any inherent explanation for their decision. This is true both for classifiers and regressors, and is especially problematic in the deep learning paradigm, where even the features are usually computed by the model itself, i.e. without human input. The lack of human-understandable features makes it difficult to infer the reasoning behind the output produced by the model. Nonetheless, several approaches to alleviate this problem have been developed.

SHAP [20] is a model-agnostic explainer (although model-specific variants also exist), meaning the explanation is not tied to the specifics of a given classifier or regressor. This is beneficial because we can thus use the same method to evaluate vastly different models. A high level overview of SHAP is the following: to determine the importance of each individual feature, we can iteratively corrupt different subsets of features (e.g. by setting them to the corresponding mean values) and observe the change in the output of the model. Based on these changes, marginal contributions are computed for each feature. Since there are exponentially many subsets of features, Monte Carlo techniques are often used for efficiency. Details of this approximation are discussed in [43].

LIME [30] is a model-agnostic explainer that uses input perturbation to train a surrogate model to locally explain individual predictions.

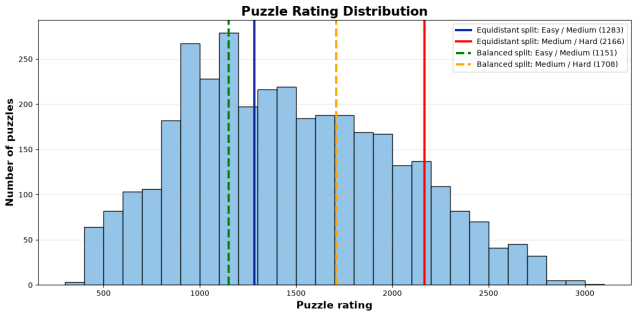


Figure 1: Rating distribution with indicated class delimiters for equidistant split (according to rating) and balanced split (according to number of samples in each class).

3 METHODOLOGY

3.1 Data

To train and evaluate our models, we use a subset of the Lichess puzzle dataset [1] containing 3500 puzzles with ratings ranging between 399 and 3050. We discard all columns except FEN, rating, and themes. Furthermore, we only keep a subset of themes which have a precise definition (for a full list, see the implementation [2]). The dataset is split into training set and test set with the ratio 80:20. The data is standardized, meaning we transform it in such a way that it follows a standard normal distribution with zero mean and unit variance. This is required by distance-based models to ensure that each feature is treated as equally important.

For classification we create classes *easy*, *medium* and *hard*. The distribution with indicated boundaries between classes, is shown in Figure 1. Classes can be constructed in two sensible ways - one by creating rating bins equidistantly, one by ensuring each bin contains the same number of samples. The latter produces a significantly narrower rating range for *middle* difficulty, which is due to the relatively low frequency of difficult puzzles in the dataset.

All experiments in Chapter 4 use balanced split, where the boundaries are determined on the train set only.

[1] https://database.lichess.org/#puzzles

[2] Full implementation is available at https://github.com/bajicluka01/chess-puzzle-evaluation/tree/main

3.2 Evaluation Metrics

For classification problem we use the standard metrics: classification accuracy, F1 score and AUC. For regression we use mean absolute error (MAE) because unlike mean squared error, it has a more intuitive interpretation on this dataset. We also report which percentage of the rating interval this MAE covers.

3.3 Features

In this section we discuss the features that were used for model training in Chapter 4.

3.3.1 Meaningful Search Tree. We use the features from [35] (except *SeeminglyGood*, which is not rigorously defined and cannot be easily reproduced) with the following modification: we generate the *MeaningfulMoves* feature using Otter [31], a recently developed engine that models human play, and use these moves to compute the remaining features from [35].

This change is motivated by the following: meaningful moves in [35] are intended to represent the moves a human player would generally consider in a given position (including moves that are quickly rejected, e.g. captures that obviously do not work). However, these moves are computed directly from the engine evaluation, which renders the feature useless in many cases, e.g. in tactical positions in which there is objectively only one good move, whereas a human player clearly needs to consider multiple alternatives. Conversely, Otter [31] is a transformer-based approach that for a given position P returns a probability distribution across all legal moves in P, indicating how likely a human player is to consider each alternative. In other words, we obtain probabilities p_i with the following properties:

$$\forall i = 1, \ldots, k: \quad p_i \geq 0, \quad \sum_{L(P)} p_i = 1 \tag{1}$$

where $L(P)$ is the set of legal moves in position P and i enumerates all the moves in $L(P)$, so $|L(P)| = k$.

In our experiments we select meaningful moves by using an arbitrarily chosen cutoff $p_i > 0.1$, which based on Equation (1) gives at most 10 (but typically less) moves most likely to be considered by a human player in a given position. In our view this coincides with the number of moves an expert player typically considers.

3.3.2 Weak Engines. The following reasoning should make intuitive sense: if a weak engine (or player) can solve a given puzzle, it is more likely to be a low-rated puzzle. Therefore we can employ several weak engines and use indicators of their puzzle-solving success as a subset of features. What exactly is meant by "weak" is not of great importance (see discussion in Chapter 4.1) - in fact it is desirable to use multiple engines with varying estimated strengths [3] to obtain less correlated features.

3.3.3 Miscellaneous. In addition to the above mentioned features, we also use Stockfish's outputs, namely centipawn evaluations, win/draw/loss rate and selective depth.

[3]Note that using, e.g. Stockfish with different playing strength parameters, gives unsatisfactory results. Evidently, making a strong engine play at less-than-full strength, is a harder (and perhaps less interesting) problem than implementing a weak engine from scratch.

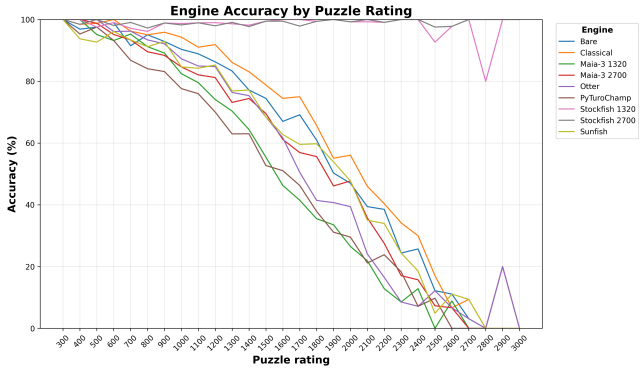


Figure 2: Engine puzzle solving success based on puzzle rating.

Furthermore, we compute the number of pins and forks in the given position based on the following definitions.

A pin occurs when a piece attacks an enemy front piece along a line (file, rank, or diagonal) behind which stands a second enemy piece. The front piece is pinned if the piece behind is worth more than the attacking piece. The front piece is pinned if the second piece is worth equal or less but is undefended.

A fork is a double attack except if one of the attacked pieces is an undefended, non-king-adjacent pawn. We also count potential forks, that is forks that can be made by any piece on the next move.

4 RESULTS

4.1 Correlation between puzzle rating and engine Elo

The quality of the Elo [1, 11] rating system in chess has been called into question numerous times, e.g. in [15, 26], however it remains a generally accepted measure of playing strength. In many settings, including Lichess (hence also in the dataset used in this paper), the Glicko-2 [13, 14] rating system is used instead [4]. On the other hand, estimation of chess engines' playing strength is generally still done using Elo. Therefore it is worth investigating the correlation between an engine's rating and its ability to solve puzzles of a similar rating.

We use the following engines: Stockfish [38] with two strength settings, 1320 and 2700, Sunfish [36], Otter [31], Classical [29], PyTuroChamp [27] and its base mode, and Maia-3 [23] with same two strength settings as Stockfish. The results are shown in Figure 2. We can observe that all of these engines except Stockfish follow the expected downward trend for increasing puzzle difficulty, therefore the puzzle solving success indicator for all except Stockfish can be used as features.

4.2 Classification

In Table 1 we show the results for the classification problem using balanced class split. Logistic regression from [25] and XGBoost [8] are used as a baseline, and are clearly outperformed by the

[4]Whether applying any rating system to an individual position rather than a player/engine even makes sense is a separate discussion, which we do not engage in here.

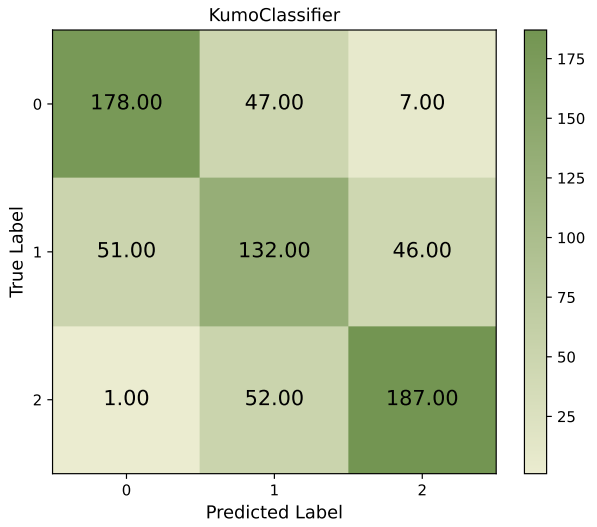


Figure 3: Confusion matrix for KumoTabular classifier.

state-of-the-art models KumoTabular [24] and TabFM [19]. Both of these are used out-of-the-box, without domain-specific fine-tuning. Ensemble refers to a soft voting scheme applied to the outputs of all other models, meaning that probabilistic predictions are aggregated and converted into a class value.

Table 1: Classification scores on 3.5k dataset.

model	CA	F1	AUC
LogReg	0.69	0.69	0.86
XGBoost [8]	0.65	0.65	0.85
Kumo-Tabular [24]	**0.70**	**0.70**	**0.87**
TabFM [19]	0.70	0.70	0.86
Ensemble	0.70	0.70	0.86

4.2.1 Misclassifications. Due to the ordinal nature of the classes and the somewhat fluid boundaries between them, it makes sense to take a closer look at misclassifications, which we display in Figure 3. As expected, strong errors (i.e. classifying an easy example as hard, or vice versa) and significantly less frequent than weak errors (i.e. easy to medium, medium to hard, or vice versa).

4.2.2 Ablation. In Table 2 we show the results of an ablation study aimed to determine the utility of each subset of features. We use the best performing model in terms of CA from the previous experiment, namely KumoTabular. Baseline refers to the following features: Stockfish centipawn evaluations, number of searched nodes for each depth, selective depth (length of the deepest principal variation) and Stockfish win/draw/loss values. Other entries in the table denote which subsets of features from Chapter 3.3 are used in addition to the baseline.

We can observe that engines have the biggest individual contribution, however they lack interpretability, so we also use other feature subsets for explanations.

Table 2: Ablation study on 3.5k dataset.

model	CA	F1	AUC
baseline	0.64	0.62	0.76
SearchTree	0.69	0.67	0.81
Engines	0.71	0.71	0.85
themes	0.66	065	0.79
Misc	0.67	0.65	0.80
SearchTree, Engines	0.73	0.73	0.86
SearchTree, themes	0.70	0.68	0.83
SearchTree, Misc	0.69	0.67	0.82
Engines, themes	0.73	0.73	0.86
Engines, Misc	0.74	0.74	0.87
themes, Misc	0.68	0.67	0.82
all features	**0.74**	**0.74**	**0.87**

4.3 Regression

In Table 3 we show the results for the regression problem. Linear regression from [25] and XGBoost [8] are used as a baseline, and are again outperformed by KumoTabular [24] and TabFM [19] without fine-tuning. Ensemble model performs unweighted averaging on the outputs of all other models.

Table 3: Regression scores on 3.5k dataset.

model	MAE	% of interval
XGBoost [8]	255.92	9.65
Linear regression [25]	250.72	9.46
Kumo-Tabular [24]	**229.47**	**8.66**
TabFM [19]	236.31	8.91
Ensemble	238.53	9.00

4.4 Explanations

In this section we discuss LIME and SHAP explanation of an arbitrarily chosen position shown in Figure 4. In both cases we explain the KumoTabular regression model.

4.4.1 LIME. Figure 5 contains the LIME explanation of the position from Figure 4. 20 most important features are visualized. The y-axis shows the feature values, the x-axis shows the amount of rating points added or subtracted from the mean for each of the features.

4.4.2 SHAP. In Figure 6 we have the SHAP explanation of the position from Figure 4. Features are again ranked based on importance.

5 CONCLUSIONS

We have surveyed the existing approaches for predicting chess puzzle difficulty, with particular emphasis on hand-crafted features that provide meaningful explanations of the underlying machine learning models. We also provided our own implementation of additional features, including puzzle solving success of weak engines.

The results on classification and regression tasks show that state-of-the-art tabular data models KumoTabular [24] and TabFM [19]

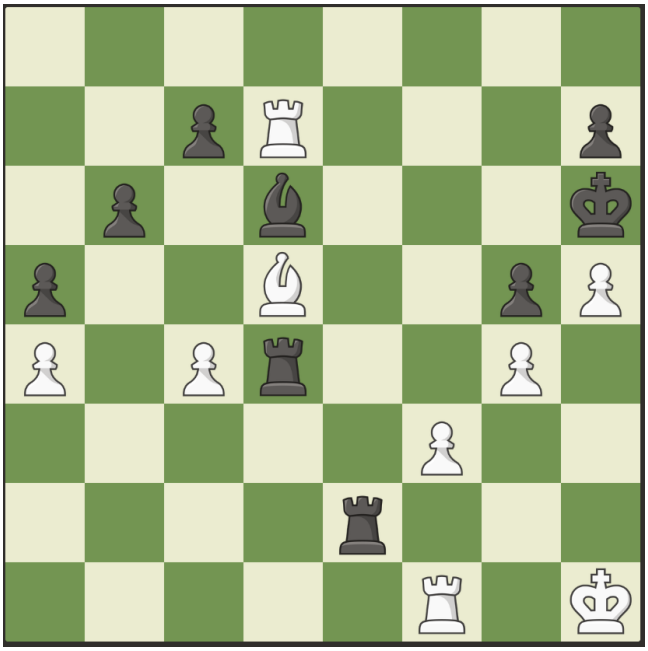


Figure 4: Chess position.

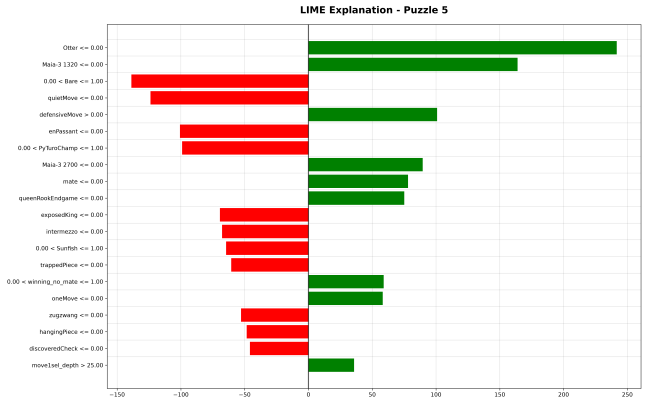


Figure 5: LIME explanation of KumoTabular regressor on the selected chess position. 20 most significant features are shown.

outperform the baseline models without any additional fine-tuning. Ablation study demonstrates the utility of weak engine features, although their meaningfulness in explainers is limited.

Future work would consist of checking if these approaches are suitable also for strategic/nondescript positions, rather than just puzzles. Here, the main challenge is in assigning the ground truth ELO to these positions, since consensus among human players is more difficult to achieve. One possible approach would be to take as difficult examples the mistakes from grandmaster games, excluding those made under time pressure. A further limitation is that none of the engine features are applicable because there is no ground truth solution.

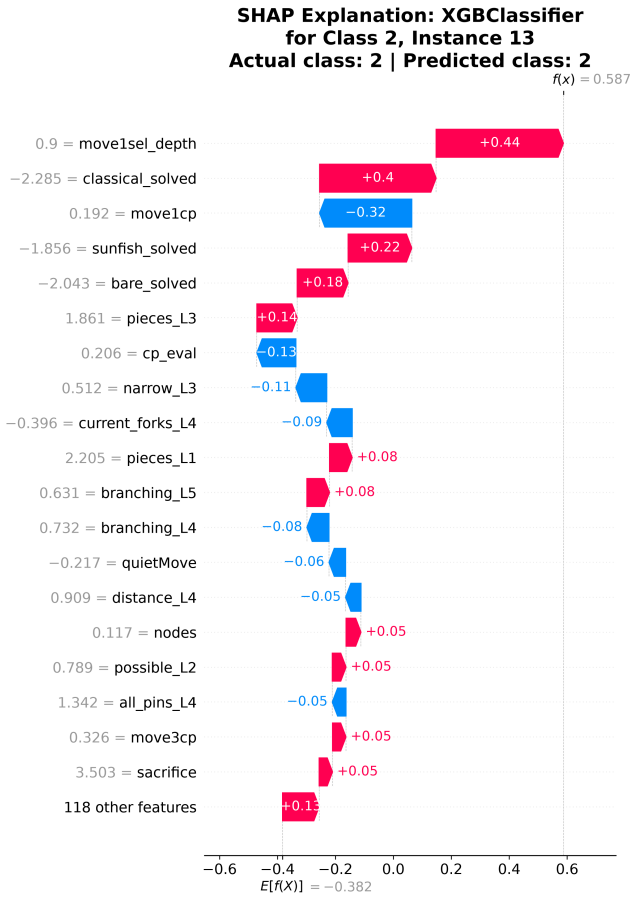


Figure 6: SHAP explanation of KumoTabular r egressor on the selected chess position. Features are ordered by importance.

Additionally, ranking approaches could also be tested. In this setting there is no need for precise rating assignment, which is a convenient property for strategic positions. However, the absence of ground truth still remains an unresolved problem.

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